
Started on Monday, 6 July 2026, 12:40 PM

State Finished

Completed on Monday, 6 July 2026, 12:47 PM

Time taken 6 mins 56 secs

Grade 6.00 out of 6.00 (100%)

Question 1

Correct

Mark 3.00 out of 3.00

Consider the following state transition probability distribution (**A**):

$$A = \{a_{ij}\} = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.6 & ? \\ 0.1 & 0.1 & 0.8 \end{bmatrix}$$

Find the missing value in the state transition matrix (**A**).

Draw the **HMM** in your paper. Then, identify its type.

Select one:

- ☐ A. Left-to-right **HMM**.
- ☒ B. Ergodic **HMM**. ✓
- ☐ C. Parallel path Left-to-right **HMM**.
- ☐ D. None of these choices.

Your answer is correct.

The correct answer is:

Ergodic **HMM**.

Question 2

Correct

Mark 1.50 out of 1.50

Match:

- The observation symbol probability distribution (**B**) represents the probability of observing the outcome given the state. ✓
- The state transition probability distribution (**A**) represents the probabilities of transitioning from one state to another. ✓
- The initial state distribution (**π**) is the probability that the Markov chain will start in a certain state. ✓

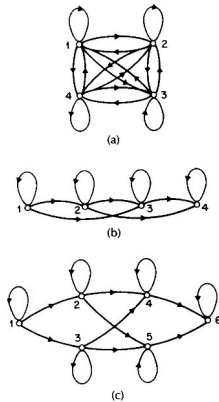
Your answer is correct.

The correct answer is:

The observation symbol probability distribution (**B**) → represents the probability of observing the outcome given the state.,The state transition probability distribution (**A**) → represents the probabilities of transitioning from one state to another.,The initial state distribution (**π**) → is the probability that the Markov chain will start in a certain state.**Question 3**

Correct

Mark 1.50 out of 1.50

Identify the type of each of the following **HMM** models:

- (a) Ergodic HMM. ✓
- (b) Left-to-right HMM. ✓
- (c) Parallel path Left-to-right HMM. ✓

Your answer is correct.

The correct answer is: (a) → Ergodic HMM.,

(b) → Left-to-right HMM.,

(c) → Parallel path Left-to-right HMM.

